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name: <unnamed>
log: C:\Dropbox\11 PS\PHD\data analysis\MDM_Resubmission.smcl
log type: smcl
opened on: 31 Dec 2024, 17:30:57

```

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1 .
2 . *** Replication for MDM paper (as of 31 December 2024)
3 .
4 . use PHD_StataData, clear
5 . describe

```

Contains data from **PHD_StataData.dta**

```

Observations:      77
Variables:         10          31 Dec 2024 09:33

```

Variable name	Storage type	Display format	Value label	Variable label
pid	byte	%8.0g		
agegrp	byte	%10.0g		
specialist	float	%9.0g		
smallgrp	float	%9.0g		
PEF1	byte	%10.0g		
PEF2	byte	%10.0g		
PEF3	byte	%10.0g		
ITR1	byte	%10.0g		
ITR2	byte	%10.0g		
ITR3	byte	%10.0g		

Sorted by:

```

6 . /*
> Contains data from PHD_StataData.dta
> Observations:      77
> Variables:         10
>
> pid                Personal identifier
> agegrp             Age group (25-29, 30-34, 35+)
> specialist         1 if specialist, 0 otherwise
> smallgrp           1 if Smaller-population group, 0 if Larger-population group
> PEF1               Perceived effectiveness in RRR (1-7)
> PEF2               Perceived effectiveness in ARR (1-7)
> PEF3               Perceived effectiveness in NNS (1-7)
> ITR1               Intention to recommend in RRR (1-7)
> ITR2               Intention to recommend in ARR (1-7)
> ITR3               Intention to recommend in NNS (1-7)
> */
7 .
8 . * Table 2: from top to bottom
9 . tabulate agegrp smallgrp, column // smallgrp = 1 if Smaller-population group, 0 othe
> rwise

```

Key

frequency
column percentage

agegrp	smallgrp		Total
	0	1	
2	19 50.00	21 53.85	40 51.95
3	18 47.37	18 46.15	36 46.75
4	1 2.63	0 0.00	1 1.30
Total	38 100.00	39 100.00	77 100.00

10. tabulate specialist smallgrp, column

Key
<i>frequency</i> <i>column percentage</i>

specialist	smallgrp		Total
	0	1	
0	31 81.58	32 82.05	63 81.82
1	7 18.42	7 17.95	14 18.18
Total	38 100.00	39 100.00	77 100.00

11.

12. * Figure 2: from left to right

13. mean PEF1 PEF2 PEF3 if smallgrp==1

Mean estimation

Number of obs = **39**

	Mean	Std. err.	[95% conf. interval]	
PEF1	4.846154	.2280542	4.384482	5.307825
PEF2	3.897436	.2539695	3.383302	4.41157
PEF3	3.615385	.2312186	3.147307	4.083462

14. mean PEF1 PEF2 PEF3 if smallgrp==0

Mean estimation

Number of obs = **38**

	Mean	Std. err.	[95% conf. interval]	
PEF1	4.5	.2606208	3.971932	5.028068
PEF2	4.394737	.2151432	3.958815	4.830658
PEF3	3.763158	.218081	3.321284	4.205032

15. mean ITR1 ITR2 ITR3 if smallgrp==1

Mean estimation

Number of obs = 39

	Mean	Std. err.	[95% conf. interval]	
ITR1	4.948718	.2517113	4.439155	5.458281
ITR2	3.871795	.2777864	3.309446	4.434144
ITR3	3.641026	.2394528	3.156279	4.125772

16. mean ITR1 ITR2 ITR3 if smallgrp==0

Mean estimation

Number of obs = 38

	Mean	Std. err.	[95% conf. interval]	
ITR1	4.894737	.234898	4.418788	5.370685
ITR2	4.236842	.248564	3.733204	4.740481
ITR3	3.684211	.2297418	3.218709	4.149712

17.

18. * Figure 3: from left to right

19. generat PEF_ARR = .

(77 missing values generated)

20. replace PEF_ARR = 1 if PEF2<=3

(24 real changes made)

21. replace PEF_ARR = 2 if PEF2==4

(19 real changes made)

22. replace PEF_ARR = 3 if PEF2>=5

(34 real changes made)

23. table PEF_ARR if smallgrp==1, stat(percent)

	Percent
PEF_ARR	
1	41.03
2	23.08
3	35.90
Total	100.00

24. table PEF_ARR if smallgrp==0, stat(percent)

	Percent
PEF_ARR	
1	21.05
2	26.32
3	52.63
Total	100.00

```

25.
26. generat ITR_ARR = .
    (77 missing values generated)

27. replace ITR_ARR = 1 if ITR2<=3
    (30 real changes made)

28. replace ITR_ARR = 2 if ITR2==4
    (12 real changes made)

29. replace ITR_ARR = 3 if ITR2>=5
    (35 real changes made)

30. table ITR_ARR if smallgrp==1, stat(percent)

```

	Percent
ITR_ARR	
1	46.15
2	15.38
3	38.46
Total	100.00

```

31. table ITR_ARR if smallgrp==0, stat(percent)

```

	Percent
ITR_ARR	
1	31.58
2	15.79
3	52.63
Total	100.00

```

32.
33. * Table 3
34. reshape long PEF ITR, i(pid) j(PRE)          // reshape the data from wide to lon
    > g
    (j = 1 2 3)

```

Data	Wide	->	Long
Number of observations	77	->	231
Number of variables	12	->	9
j variable (3 values)		->	PRE
xij variables:			
	PEF1 PEF2 PEF3	->	PEF
	ITR1 ITR2 ITR3	->	ITR

```

35. generate smallARR=(smallgrp==1 & PRE==2)      // interaction term

```

```

36.
37. xtreg PEF i.PRE smallARR i.agegrp i.specialist, re i(pid)

```

Random-effects GLS regression	Number of obs	=	231
Group variable: pid	Number of groups	=	77
R-squared:	Obs per group:		
Within = 0.2172	min =		3
Between = 0.0051	avg =		3.0
Overall = 0.0830	max =		3
corr(u_i, X) = 0 (assumed)	Wald chi2(6)	=	42.36
	Prob > chi2	=	0.0000

PEF	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PRE						
2	-.2468747	.2099717	-1.18	0.240	-.6584116	.1646623
3	-.987013	.1613067	-6.12	0.000	-1.303168	-.6708577
smallARR	-.5638629	.2653898	-2.12	0.034	-1.084017	-.0437084
agegrp						
3	-.0721906	.3221041	-0.22	0.823	-.703503	.5591218
4	.4419878	1.318251	0.34	0.737	-2.141736	3.025712
1.specialist	-.0823305	.4271039	-0.19	0.847	-.9194387	.7547777
_cons	4.718305	.2157477	21.87	0.000	4.295447	5.141163
sigma_u	1.088798					
sigma_e	1.0028114					
rho	.54104092	(fraction of variance due to u_i)				

38. est store re

39. xtreg PEF i.PRE smallARR, fe i(pid)

Fixed-effects (within) regression	Number of obs	=	231
Group variable: pid	Number of groups	=	77
R-squared:	Obs per group:		
Within = 0.2173	min =		3
Between = 0.0017	avg =		3.0
Overall = 0.0804	max =		3
corr(u_i, Xb) = -0.0094	F(3,151)	=	13.97
	Prob > F	=	0.0000

PEF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
PRE						
2	-.2303486	.2150024	-1.07	0.286	-.65515	.1944528
3	-.987013	.1616177	-6.11	0.000	-1.306337	-.6676889
smallARR	-.5964912	.2799537	-2.13	0.035	-1.149623	-.0433591
_cons	4.675325	.114281	40.91	0.000	4.449528	4.901121
sigma_u	1.2038142					
sigma_e	1.0028114					
rho	.59034096	(fraction of variance due to u_i)				

F test that all u_i=0: F(76, 151) = **4.32** Prob > F = **0.0000**

40. est store fe

41. hausman fe re, sigmamore // RE favored

Note: the rank of the differenced variance matrix (**1**) does not equal the number of coefficients being tested (**3**); be sure this is what you expect, or there may be problems computing the test. Examine the output of your estimators for anything unexpected and possibly consider scaling your variables so that the coefficients are on a similar scale.

	Coefficients			
	(b) fe	(B) re	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
PRE				
2	-.2303486	-.2468747	.0165261	.044274
3	-.987013	-.987013	5.00e-15	.
smallARR	-.5964912	-.5638629	-.0326284	.0874127

b = Consistent under H0 and Ha; obtained from **xtreg**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

chi2(1) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = 0.14
 Prob > chi2 = 0.7089
 (V_b-V_B is not positive definite)

42.

43. xtreg ITR i.PRE smallARR i.agegrp i.specialist, re i(pid)

Random-effects GLS regression Number of obs = 231
 Group variable: **pid** Number of groups = 77

R-squared: Obs per group:

Within = 0.3010	min = 3
Between = 0.0147	avg = 3.0
Overall = 0.1192	max = 3

corr(u_i, X) = 0 (assumed) Wald chi2(6) = 66.39
 Prob > chi2 = 0.0000

ITR	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PRE						
2	-.6874753	.2110976	-3.26	0.001	-1.101219	-.2737316
3	-1.25974	.1617891	-7.79	0.000	-1.576841	-.9426395
smallARR	-.3606256	.2677169	-1.35	0.178	-.885341	.1640898
agegrp						
3	-.0070677	.3437472	-0.02	0.984	-.6807999	.6666645
4	1.235897	1.406725	0.88	0.380	-1.521233	3.993027
1.specialist	-.1990064	.4558025	-0.44	0.662	-1.092363	.6943501
_cons	4.945515	.2277283	21.72	0.000	4.499175	5.391854
sigma_u	1.1813171					
sigma_e	1.0060915					
rho	.57959571	(fraction of variance due to u_i)				

44. est store re

45. xtreg ITR i.PRE smallARR, fe i(pid)

Fixed-effects (within) regression Number of obs = 231
 Group variable: **pid** Number of groups = 77

R-squared: Obs per group:

Within = 0.3010	min = 3
Between = 0.0021	avg = 3.0
Overall = 0.1113	max = 3

corr(u_i, Xb) = -0.0002 F(3,151) = 21.68
 Prob > F = 0.0000

ITR	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
PRE						
2	-.6825017	.2157056	-3.16	0.002	-1.108693	-.2563108
3	-1.25974	.1621463	-7.77	0.000	-1.580109	-.9393717
smallARR	-.3704453	.2808694	-1.32	0.189	-.9253867	.1844961
_cons	4.922078	.1146548	42.93	0.000	4.695543	5.148613
sigma_u	1.2895259					
sigma_e	1.0060915					
rho	.62161369	(fraction of variance due to u_i)				

F test that all u_i=0: F(76, 151) = 4.93 Prob > F = 0.0000

46. est store fe

47. hausman fe re, sigmamore // RE favored

Note: the rank of the differenced variance matrix (1) does not equal the number of coefficients being tested (3); be sure this is what you expect, or there may be problems computing the test. Examine the output of your estimators for anything unexpected and possibly consider scaling your variables so that the coefficients are on a similar scale.

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) fe	(B) re		
PRE				
2	-.6825017	-.6874753	.0049736	.0419749
3	-1.25974	-1.25974	2.22e-15	.
smallARR	-.3704453	-.3606256	-.0098197	.0828736

b = Consistent under H0 and Ha; obtained from **xtreg**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

chi2(1) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= 0.01
Prob > chi2 = 0.9057
(V_b-V_B is not positive definite)

48.

49. xtreg PEF i.PRE smallARR smallgrp i.agegrp i.specialist, re i(pid)

Random-effects GLS regression Number of obs = 231
Group variable: pid Number of groups = 77

R-squared: Obs per group:

Within = 0.2173	min = 3
Between = 0.0059	avg = 3.0
Overall = 0.0841	max = 3

corr(u_i, X) = 0 (assumed) Wald chi2(7) = 42.34
Prob > chi2 = 0.0000

PEF	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PRE						
2	-.2303486	.2150024	-1.07	0.284	-.6517455	.1910483
3	-.987013	.1616177	-6.11	0.000	-1.303778	-.6702481
smallARR	-.5964912	.2799537	-2.13	0.033	-1.14519	-.0477921
smallgrp	.1111281	.2982908	0.37	0.709	-.4735111	.6957674
agegrp						
3	-.0675049	.32297	-0.21	0.834	-.7005146	.5655047
4	.5006554	1.330147	0.38	0.707	-2.106385	3.107696

1.specialist	-.0883657	.4282338	-0.21	0.837	-.9276886	.7509571
_cons	4.660164	.2666125	17.48	0.000	4.137613	5.182715
sigma_u	1.088798					
sigma_e	1.0028114					
rho	.54104092	(fraction of variance due to u_i)				

50. outreg2 using table3.doc, replace ci dec(3) excel 2aster

table3.doc
table3.xml
 dir : seeout

51. xtreg ITR i.PRE smallARR smallgrp i.agegrp i.specialist, re i(pid)

Random-effects GLS regression	Number of obs	=	231
Group variable: pid	Number of groups	=	77
R-squared:	Obs per group:		
Within = 0.3010	min =		3
Between = 0.0148	avg =		3.0
Overall = 0.1194	max =		3
	Wald chi2(7)	=	66.11
corr(u_i, X) = 0 (assumed)	Prob > chi2	=	0.0000

ITR	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PRE						
2	-.6825017	.2157056	-3.16	0.002	-1.105277	-.2597265
3	-1.25974	.1621463	-7.77	0.000	-1.577541	-.9419393
smallARR	-.3704453	.2808694	-1.32	0.187	-.9209392	.1800485
smallgrp	.0374316	.3166018	0.12	0.906	-.5830965	.6579598
agegrp						
3	-.0054712	.3447709	-0.02	0.987	-.6812097	.6702673
4	1.255886	1.419933	0.88	0.376	-1.527132	4.038905
1.specialist	-.2010628	.45714	-0.44	0.660	-1.097041	.6949152
_cons	4.925924	.282041	17.47	0.000	4.373133	5.478714
sigma_u	1.1813171					
sigma_e	1.0060915					
rho	.57959571	(fraction of variance due to u_i)				

52. outreg2 using table3.doc, append ci dec(3) excel 2aster

table3.doc
table3.xml
 dir : seeout

53.

54. log close
 name: <unnamed>
 log: C:\Dropbox\11 PS\PHD\data analysis\MDM_Resubmission.smcl
 log type: smcl
 closed on: 31 Dec 2024, 17:30:58